13 Protecting your application against external attacks

This chapter covers

- Configuring your application for HTTPS
- Ensuring you apply good access control
- Avoiding common cryptographic failures
- Preventing injection attacks

Once your application is published, it will come under attack. Most of these attacks will be automated, probing for well-known vulnerabilities. You will probably be surprised to see requests being logged for URLs that include *phpmyadmin*, or the names of popular third-party component libraries you don't even use. The HTTPS protocol underpins the security of any website. We will begin this chapter with a reminder of the importance of HTTPS and learning how to configure your application to use HTTPS.

Then we will look at the threats posed to your application in more detail, focusing on the most common of these according to the Open Web Application Security Project (OWASP) (https://owasp.org/). You will learn how the Razor Pages framework has been designed specifically to protect you from various threats and how to maintain security should you find the need to bypass these safeguards.

The OWASP categorizes vulnerabilities under a number of headings. We will look at some of the specific threats under the top three headings:

- Broken access control
- Cryptographic failures
- Injection attacks

We will examine vulnerabilities grouped under the broken access control heading, which includes issues affecting authentication and authorization. Then, you will learn about some common cryptographic failures, such as storing passwords in plain text, and how to avoid